Noise Removal in Transmission of Quantum Electromagnetic Signals Through Optical Fibers: Computation of Maximum Bit Rate Using Classical-Quantum (Cq) Channel Capacity

Kumar Gautam[1](IEEE, Senior Member) [1]Founder and CEO, Quantum Research And Centre of Excellence, Delhi, India.

Corresponding Author: [1]***Kumar Gautam, Email: edumonk@ieee.org***

Abstract

In this work, quantum information theory is utilized to address noise removal in the transmission of quantum electromagnetic signals through optical fibres. The study figures out the fastest reliable bit rate for sending classical data by using the classical capacity of quantum channels, more specifically the Holevo capacity. It demonstrates that traditional post-channel noise correction methods, utilizing completely positive trace-preserving (CPTP) maps, inadvertently lower capacity due to the data processing inequality. However, incorporating noise mitigation into system dynamics by using non-commuting control superoperators, on the other hand, allows for an effective correction map that is trace-preserving but not completely positive, enabling simultaneous noise reduction and increased bit rate. The information transmission can be examined from two perspectives: the Heisenberg picture, where the state of the field is fixed while observables evolve, and the Schrödinger picture, where observables are fixed and the state changes over time. The preferred Schrödinger picture facilitates the assessment of quantum expectations of all spatial field moments. The process begins with an initial state comprised of the system state and a bath, evolving via a unitary transformation defined by a perturbed Hamiltonian. In a binary message with classical information encoding, there are two different quantum states for each bit. The Holevo-Schumacher-Westmoreland theorem establishes that the mutual information, when aligned with the Holevo quantity, provides a maximum reliable rate, making it possible to determine the maximum bit rate accordingly. Specific commonly used thermal states are referenced for encoding during the transmission process. The mathematical model represents the quantized electromagnetic field that travels through the optical fibre, and the noise-induced dynamics are contained in a Gorini-Kossakowski-Sudarshan-Lindblad equation. A composite mapping is produced when a post-channel correction map is attempted to be applied, and it is demonstrated that this cannot improve capacity because of the data processing inequality. The research flips the script and explores integrated noise management by adding a control superoperator to the master equation. This updated map outperforms conventional noise channels in terms of effective capacity due to non-commuting dynamics. This demonstrates that integrated control can increase the maximum reliable bit rate and decrease noise. The results show how important noise-free control dynamics are because they can expand capacity a lot beyond what would be possible without control. This innovative approach highlights the potential for significant advancements in communication systems by leveraging integrated noise management techniques. By effectively manipulating control dynamics, researchers have shown promising results in enhancing data transmission capabilities and minimizing interference from noise.

Index Terms

Schrödinger equation, Quantum mechanical time independent Perturbation theory, Time averaging, Brownian motion, Ito calculus, Quantum parameter estimation theory

I. Introduction

Theoretical framework for quantum communication in which the basic Cq communication problem is stated, with classical alphabets encoded into states and detection operators used to decode the input alphabet string from the tensor product state output. By applying the Cq Shannon capacity theorem, we can determine the highest rate of reliable information transmission across this channel. This calculation relies on the mixed quantum state's Von-Neumann quantum entropy. In this real-world quantum problem, the Cq communication theory is applied by means of electromagnetic fields produced by current sources, where the receiver is an electron bound to an atomic nucleus, which interacts with the received field and, after a finite amount of time, changes its state to a value that is dependent on the current sources that are transmitting it. We conclude by discussing certain aspects of real-time nonlinear stochastic filtering theory and its potential uses in estimating medium and antenna parameters from noisy electromagnetic field measurements taken at discrete locations.

Quantum communication, utilizing methods like teleportation and super-dense coding, surpasses classical communication in speed and efficiency. In teleportation, sharing a maximally entangled state allows for the transmission of log2(d)-qubits of quantum information by sending log2(2d) classical bits. While classical bits transmit only ones or zeros, qubits require three real numbers and, through entanglement, can effectively be transmitted at infinite speed. This is illustrated by the EPR paradox, where the measurement of one entangled particle instantaneously influences the outcome of another, regardless of distance.

However, classical communication is still necessary for reconstructing qubits, with two classical bits needed to indicate the superpositions of one teleported qubit, as ebits alone cannot transmit information without a classical channel.

In spectrum analysis, traditional fast Fourier transform (FFT) algorithms with a complexity of $O(N \log N)$ can be outperformed by the quantum Fourier transform that operates at $O(N)$. Quantum communication, utilizing quantum teleportation and superdense coding, allows for faster transmission of data by leveraging entangled states. The principles behind this include the EPR paradox, facilitating instant transmission across distances. In measuring quantum observables like spin and momentum, conventional filtering struggles due to Heisenberg uncertainty, necessitating more advanced quantum filtering methods. Quantum computers are expected to replace classical ones, utilizing quantum registers for information storage and requiring non-demolition measurements for accurate data assessment. Additionally, energy levels in quantum systems can be computed using Hamiltonian operators and Schrodinger's unitary operator. The quantum Fourier transform plays a crucial role in estimating phases related to energy levels in various physical systems, including semiconductors. By the completion of the quantum communication process, we want the receiver to have returned the original state ρ. Consider the scenario where an atom is excited in the transmitter's electromagnetic field, is moved around by the message, and its present status is transmitted across a channel. By recovering at the receiving end, we may determine that the message's electromagnetic field was precisely on target. In the midst of the chaotic channel, one gets a state from the receiver that closely resembles the original state of the atom in the transmitter's field, indicating successful quantum communication. This demonstrates the effectiveness of transmitting quantum information through a channel despite potential disturbances:

$$\mathfrak{k}_1(\rho) = \sum_{m=1}^{p} E_m \rho E_m^*$$

and at the receiving end, we need to determine recovery operators $\{R_k\}$ that depend only on the channel noise operators $\{E_m\}$, so that the output state of the receiver

$$\mathfrak{k}_2 o \mathfrak{k}_1(\rho) = \sum_m R_m \mathfrak{k}_1(\rho) R_m^* = \rho$$

This perfect recovery is possible only if the channel noise operators $\{E_m\}$ and the range space of the transmitted state ρ satisfy the conditions of the conditions of the Knill-Laflamme theorem. In this case, the design of the recovery operators $\{R_m\}$ can be reduced to a quantum gate design problem by using Stinespring/Choi-Kraus representation of a quantum operation in terms of a reference state, a unitary gate and a measurement involving partial trace. The design of the unitary gate can be achieved using physical systems. Further, the simulation of the quantum channel $\mathfrak{k}_1 = \{E_m\}$ can also be accomplished using unitary gate design in the Choi-Kraus representation:

$$\|_1(\rho) = Tr_2(U(\rho \otimes \rho_0)U^*)$$

In Cq communication, we wish to transmit a string of N statistical independent symbols from an alphabet A with each symbol x having a probability $p(x)$ associated with it. We associate a quantum state $\rho(x)$ with each symbol x in the alphabet. Thus, in order to transmit the string $(x_1 x_2 ... x_N)$ we transmit the quantum state $\rho_N(x_1, ..., x_N) = \rho(x_1) \otimes ... \otimes \rho(x_N)$. Assuming the channel to be noiseless, at the receiver end, we wish to decode the string $(x_1, x_2, ..., x_N)$ from the received state ρ_N with very small error probability. The Cq coding theorem proved by Winter and Holevo states that, for all sufficiently large N, we can select M_N distinct source alphabet strings $u_1, ..., u_N$ of length N so that after these strings are encoded into quantum states as above, then detection operators $D_1, ..., D_M$ can be constructed at the receiver end, so that the correct detection probabilities $Tr(\rho_N(u_k)D_k), k = 1, 2, ..., M$ are arbitrarily close to unity (ie, converge to unity as $N \to \infty$) provided that $lim_{N\to\infty} log(M_N)/N \leq C$ where C is the Cq channel capacity defined by

$$C = C(\rho) = max_{\{p(x)\}}[S(\sum_x p(x)\rho(x)) - \sum_x p(x) S(\rho(x))]$$

The design of the detection operators $D_k, k = 1, 2, ..., M$ can be achieved by reducing POVM measurements to PVM measurements and then implementing the PVM measurements using the Choi-Kraus representation based on design of a unitary matrix and a partial trace operation in which the unitary matrix can be designed using quantum physical systems.

II. Problem Formulation

It is well known that when two persons share and entangled state, then quantum teleportation is possible, ie a quantum state can be transmitted between them by transmitting just a few classical bits. This is a generalization of the EPR paradox which states that if two persons share each have a spin $1/2$ particle and are initially very close to each other so that by the Pauli exclusion principle, one has spin component $+1/2$ while the other has spin component $-1/2$, then after they get separated by a considerable distance, if one of them flips the spin of his particle, then simultaneously in order to conserve the spin angular momentum, the spin of the other person will also get flipped and so he will know that his partner has flipped his particle's spin. This is in contradiction with the fundamental principles of relativity that energy and information cannot be transmitted

at a speed greater than the speed of light. However, today we know that this EPR paradox is not really a paradox, indeed entanglement enables us to transmit information at infinite speed. This is a fundamental law of quantum mechanic and it does violate the causality principle of special relativity.

In Cq quantum communication, A has a classical alphabet $\{x : x \in A\}$ and he wishes to transmit a string of his alphabets $(x_1...x_n)$ to B through a quantum channel. For this he encodes each alphabet $x \in A$ into a density matrix $\rho(x)$ all defined on a given finite dimensional Hilbert space. The alphabet x is emitted with a probability $p(x)$. So if there is no channel, B receives the state $\sum_{x\in A} p(x)\rho(x) = \bar{\rho}$ if A transmits a single alphabet. If A transmits an n-length string, then the state received by B is

$$\sum_{x_1,...,x_n \in A} p(x_1)...p(x_n) \otimes_{k=1}^{n} \rho(x_k)$$

$$= \bar{\rho}^{\otimes n}$$

The Cq-Shannon coding theorem proved by Holevo and Winter states that if

$$C = max_P(H(\sum_{x\in A} \rho(x)) - \sum_{x\in A} P(x)H(\rho(x)))$$

then for all sufficiently large n, there exist M_n distinct input strings $u_1^{(n)}, ..., u_{M_n}^{(n)} \in A^n$ and positive definite detection operators $D_1^{(n)}, ..., D_{M_n}^{(n)}$ in $\mathcal{H}^{\otimes n}$ such that $D_1^{(n)} + ... + D_{M_n}^{(n)} = I$ and

$$liminf_{n\to\infty} log(M_n)/n = C$$

with simultaneously

$$lim_{n\to\infty} min_{1\le k\le M_n} Tr(\rho^{\otimes n}(u_k^{(n)})D_k^{(n)}) = 1$$

and further, if for all sufficiently large n, we choose M_n distinct sequences $u_1^{(n)}, ..., u_{M_n}^{(n)}$ in A^n and any set $D_1^{(n)}, ..., D_{M_n}^{(n)}$ of positive definite operators such that $D_1^{(n)} + ... + D_{M_n}^{(n)} = I$, then if

$$liminf_{n\to\infty} log(M_n)/n > C$$

then

$$lim_{n\to\infty} min_{1\le k\le M_n} Tr(\rho^{\otimes n}(u_k^{(n)} D_k^{(n)}) = 0$$

This means that asymptotically zero probability of error decoding is possible if the rate of information transmission is smaller than the Cq capacity C and conversely if the rate of information transmission is larger than the Cq capacity C then asymptotically any detection scheme will produce a decoding error probability of one.

A typical example of quantum communication using a physical system is as follows: An electromagnetic field generated by a set of message currents defined by the parameter vector θ is to be communicated. The electromagnetic field generated by the current parameter θ is obtained using the classical retarded potential formula. We write this as $A(t, r|\theta), \Phi(t, r|\theta)$. This em field gets transmitted through a channel and at the receiver end, it is incident upon an atom with an electron, thereby causing the total Hamiltonian of the atom and its interaction with this em radiation to be given by

$$H(t|\theta) = (\mathbf{p} + e\mathbf{A}(t, r|\theta))^2/2m + V(r) - e\Phi(t, r|\theta)$$

This Hamiltonian causes the unitary evolution operator of the atom to evolve according to Schrodinger's equation:

$$idU(t|\theta)/dt = H(t|\theta)U(t|\theta)$$

We write

$$H(t|\theta) = H_0 + V(t|\theta)$$

where

$$H_0 = \mathbf{p}^2/2m + V(r)$$

is the atomic Hamiltonian and

$$V(t|\theta) = e^2A^2(t, r|\theta)/2m + (e/m)((p, A(t, r|\theta)) + (A(t, r|\theta), p)) - e\Phi(t, r|\theta)$$

with

$$p = -i\nabla_r$$

The unitary evolution operator for the atom is then obtained using the Dyson series:

$$U(t|\theta) = U_0(t)W(t|\theta), U_0(t) = exp(-itH_0),$$

$$W(t|\theta) = I + \sum_{n\geq 1}(-i)^n \int_{0<t_n<...<t_1<t} V(t_1|\theta)...V(t_n|\theta)dt_1...dt_n$$

The state $\rho(0)$ of the atom at time $t = 0$, after time $t = T$ evolves to

$$\rho_T(\theta) = U(T|\theta)\rho(0)U(T|\theta)^*$$

The problem of estimating θ from the measured final atomic state $\rho_T(\theta)$ is then the of the Cq-communication type. We may also consider the situation when the current parameter θ has a probability distribution $p(\theta|\alpha)$ dependent upon another parameter α. The communication problem is then to estimate the transmitted message α from the received state

$$\bar{\rho}_T = \int \rho_T(\theta)p(\theta|\alpha)d\theta$$

We may also consider the situation in which the electromagnetic field generated by the currents is a quantum electromagnetic field described by photon creation and annihilation operators. in that case, we assume that the electromagnetic field bath is in a coherent state $|\phi(u) >$ on which the creation and annihilation operators of the field act as

$$a_k|\phi(u) >= u_k|\phi(u) >, a_k^*|\phi(u) >= (\partial/\partial u_k + u_k)|\phi(u) >$$

The operator $W(T|\theta)$ and hence $U(T|\theta)$ will now act on the system$\otimes$ bath space resulting in the final joint state being given by

$$\rho(\theta) = U(T|\theta)(\rho_s(0) \otimes |\phi(u) >< \phi(u)|)U(T|\theta)^*$$

and the system state alone is given by tracing out over the bath state:

$$\rho_s(\theta) = Tr_2(\rho(\theta))$$

Now the problem again reduces to the Cq-problem of estimating θ from $\rho(\theta)$.

Another way of estimating the current parameter vector θ is based on taking measurements at discrete times $0 < t_1 < t_2 < ... < t_n$ on the quantum system, each time the noting the measurement outcome causing the state of the system to collapse. If for example, $\rho(0)$ is the initial system state, and $\mathbf{M} = \{M_a : a = 1, 2, .., N\}$ is a POVM (Positive operator valued measurement, ie, $\sum_{a=1}^N M_a^* M_a = I$), then the probability that this measurement taken at times $t_1, ..., t_n$ will give the outcomes $a_1, ..., a_n$ is

$$P_{\mathbf{M}}(a_1, ..., a_n; t_1, ..., t_n|\theta) =$$

$$Tr(M_{a_n}U(t_n, t_{n-1}|\theta)M_{a_{n-1}}U(t_{n-1}, t_{n-2}|\theta)M_{a_{n-2}}...U(t_2, t_1|\theta)M_{a_1}U(t_1, 0|\theta)\rho(0).$$

$$.U(t_1, 0|\theta)^* M_{a_1}^* U(t_2, t_1|\theta)^{\cdot}..U(t_n, t_{n-1}|\theta)^* M_{a_n}^*)$$

After noting the measurements $a_1, ..., a_n$, we maximize this probability w.r.t θ to obtain the maximum likelihood parameter estimates. The quantum Cramer-Rao lower bound on the variance matrix of the parameter estimation error taking into account all possible systems of measurements $\mathbf{M}$ applied to the system at times $t_1, ..., t_n$ is given by

$$\mathbb{E} \parallel \hat{\theta} - \theta \parallel^2 \geq min_{\mathbf{M}} Tr(J_{\mathbf{M}}(\theta)^{-1})$$

where $J_{\mathbf{M}}$ is the classical Fisher information matrix for the measurement $\mathbf{M}$:

$$J_{\mathbf{M}}(\theta) = -\sum_{a_1,...,a_n} P_{\mathbf{M}}(a_1, ..., a_n; t_1, ..., t_n|\theta)\frac{\partial^2 log(P_{\mathbf{M}}(a_1, ..., a_n; t_1, ..., t_n|\theta)}{\partial\theta\partial\theta^T}$$

Quantum Error Correcting Codes: If $\mathcal{H}$ is a finite dimensional Hilbert space, $\mathcal{C}$ is a subspace of $\mathcal{H}$ and $\mathcal{N}$ is a linear manifold of operators in $\mathcal{H}$, then we say that the code subspace $\mathcal{C}$ corrects the noise manifold $\mathcal{N}$ if given any state ρ ($\rho \geq 0, Tr(\rho) = 1$) with range contained in $\mathcal{C}$ and operators $E_1, ..., E_p \in \mathcal{N}$ such that $\sum_{k=1}^p E_k^* E_k = I$, there exist operator ("recovery operators") $R_1, ..., R_m$ in $\mathcal{H}$ depending only on $E_1, ..., E_p$ and $\mathcal{C}$ and not on ρ such that

$$RE(\rho) = \lambda\rho$$

where λ is some scalar. Here $E(.)$ is the quantum noisy channel defined by

$$E(\rho) = \sum_{k=1}^p E_k\rho E_k^*$$

and $R(.)$ is the recovery/noise removing channel

$$R(\rho) = \sum_{k=1}^m R_k\rho R_k^*, \sum_k R_k^* R_k = I$$

For a given code subspace $\mathcal{C}$ and noise operator manifold $\mathcal{N}$, the Knill-Laflamme theorem says that $\mathcal{C}$ corrects $\mathcal{N}$ (ie there exist recovery operators) iff

$$PN_2^*N_1P = \lambda(N_2^*N_1)P$$

for all $N_1, N_2 \in \mathcal{N}$ and where P is the orthogonal projection onto $\mathcal{C}$. $\lambda(N_2^*N_1)$ is some scalar dependent on $N_2^*N_1$. An intuitive proof of this theorem is based on taking $|\psi >\in \mathcal{C}$ and $E_1, ..., E_p \in \mathcal{N}$ and noting that if $\{R_k\}$ are recovery operators, then

$$\sum_{k,m} R_kE_m|\psi >< \psi|E_m^*R_k^* = \lambda|\psi >< \psi|$$

implies that for all $|\phi >$ orthogonal to $|\psi >$, we have

$$\sum_{k,m} | < \phi|R_kE_m|\psi > |^2 = 0$$

ie

$$< \phi|R_kE_m|\psi >= 0 \forall k, m$$

which implies that

$$R_kE_m|\psi >= \mu_{km}|\psi >, \mu_{km} \in \mathbb{C}$$

ie, the recovery operator R_k removes the effect of the noise operator E_m. it then follows that since $|\psi >$ is any vector in $\mathcal{C}$, we have

$$R_kE_mP = \lambda_{km}P$$

and hence

$$PE_j^*E_mP = \sum_k PE_j^*R_k^*R_kE_mP = (\sum_k \lambda_{km}\bar{\lambda}_{kj})P$$

which is the essential content of the Knill-Laflamme formula. For the full mathematical details regarding the construction of the recovery operators R_k from P and $\mathcal{N}$, we refer to [9].

III. QUANTUM CHANNEL ESTIMATION

Let $U_1, ..., U_n$ be n unitary matrices and $p_1, .., p_n$ be non-negative numbers adding to unity. Then

$$\rho \to T(\rho) = \sum_{k=1}^{n} p_kU_k\rho U_k^*$$

defines a quantum channel, ie, T is a completely positive trace preserving (CPTP) map. Suppose ρ_1 and ρ_2 are two mixed states and we wish to determine unitary matrices $U_1, ..., U_n$ such that

$$\sum_{m=1}^{N} (Tr(\rho_2Y_m) - Tr(T(\rho_1)Y_m))^2$$

is a minimum for given observables $Y_m, m = 1, 2, ..., N$. If δH is an infinitesimal Hermitian operator and U a unitary operator, then $(1 + i\delta H)U$ us unitary upto first order. Using, this we deduce that the above error energy is a minimum when

$$\sum_m (Tr(\rho_2Y_m) - Tr(T(\rho_1)Y_m))Tr(Y_m\delta T(\rho_1)) = 0$$

where

$$\delta T(\rho) = i\sum_k p_k[\delta H_k, U_k\rho U_k^*]$$

with δH_k being arbitrary infinitesimal Hermitian operators. Equating to zero the coefficient of δH_k gives

$$\sum_m (Tr(\rho_2Y_m) - Tr(T(\rho_1)Y_m))p_kTr[Y_m(\delta H_kU_k\rho_1U_k^* - U_k\rho_1U_k^*\delta H_k)] = 0$$

from which we deduce using $Tr(AB) = Tr(BA)$ that

$$\sum_{m,k} p_k(Tr(\rho_2Y_m) - Tr(T(\rho_1)Y_m))[U_k\rho_1U_k^*, Y_m] = 0$$

This is a fourth degree polynomial equation in $U_k, U_k^*, k = 1, 2, ..., n$ and can be solved only by numerical methods.

We assume now that the $U_k's$ are known and they suffer small multiplicative perturbations $1 + i\delta H_k$. We wish to estimate the perturbations δH_k subject to knowledge about the perturbed output density matrix $\rho_2 + \delta\rho_2$. A straightforward linearization argument gives

$$\delta\rho_2 = \sum_k p_k[\delta U_k \rho_1 U_k^* + U_k \rho_1 \delta U_k^*]$$
$$= i\sum_k p_k[\delta H_k, U_k \rho_1 U_k^*]$$

and we estimate δH_k so that

$$E(\{\delta H_k\}) = \sum_m (Tr(\delta\rho_2 Y_m) - i\sum_k p_k Tr([\delta H_k, U_k\rho_1 U_k^*]Y_m)^2$$

is a minimum. For convenience of notation, we denote δH_k by H_k and then δH_k will denote the perturbation in H_k (ie, the perturbation in the original δH_k). Thus, we have to minimize

$$E(\{H_k\}) = \sum_m (b[m] - i\sum_k p_k Tr([H_k, F_k]Y_m))^2$$

where

$$b[m] = Tr(\delta\rho_2 Y_m), F_k = U_k \rho_1 U_k^*$$

Setting the variational derivative of E w.r.t H_r to zero gives

$$\sum_m (b[m] - i\sum_k p_k Tr([H_k, F_k]Y_m))^2 Tr([\delta H_r, F_r]Y_m) = 0$$

which gives on using the arbitrariness of the Hermitian perturbation δH_r,

$$\sum_m (b[m] - i\sum_k p_k Tr([H_k, F_k]Y_m))[F_r, Y_m] = 0$$

or equivalently,

$$\sum_m b[m][F_r, Y_m] = i\sum_k Tr([H_k, F_k]Y_m)[F_r, Y_m], 1 \leq r \leq$$

These are linear equations for the optimum perturbations $\{H_r\}$ of the channel.

IV. Conclusions

This work has applied quantum information theory to the fundamental problem of transmitting classical information encoded in quantum electromagnetic signals through noisy optical fibers. By modeling the channel using the Schrödinger picture and the Lindblad master equation, we computed the maximum reliable bit rate as the Holevo capacity C divided by the per-symbol transmission time τ. A key insight is that conventional post-channel noise removal implemented via completely positive trace-preserving (CPTP) maps inevitably reduces the effective classical capacity due to the monotonicity of quantum relative entropy (the data processing inequality). In contrast, integrated noise mitigation—achieved by adding a control superoperator θ_2 to the original noisy dynamics θ_1—results in a modified evolution $S_t = \exp(t(\theta_1+\theta_2))$. When θ_1 and θ_2 do not commute, the effective correction map K_t is trace-preserving but not necessarily completely positive, circumventing the data processing inequality. This opens the possibility of simultaneously reducing noise and increasing the achievable bit rate beyond the uncontrolled Holevo capacity C/τ. By embedding noise mitigation directly into the system Hamiltonian or Lindblad operators, rather than applying it as a post-processing step, significant gains in reliable transmission rates become theoretically attainable. Future work should focus on designing specific non-commuting control Hamiltonians and Lindblad terms tailored to realistic optical fiber noise models (e.g., amplitude damping, phase noise, and thermal disturbances), followed by numerical simulations and experimental validation using platforms such as few-mode fibers and homodyne detection. Ultimately, this approach paves the way for next-generation quantum-enhanced classical communication systems, offering higher data rates and greater robustness in practical optical networks while remaining fully compatible with existing fiber infrastructure.

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